CLINICAL · NECK, THYROID & SALIVARY

Head and neck cancer in children

Most lumps in a child's neck are reactive nodes, and cancer is rare. But when it comes, it is not the squamous carcinoma of adults: it is lymphoma, a sarcoma, a neural tumour or a thyroid cancer, and diagnostic delay is common. The ENT surgeon's job is to recognise the lump that does not behave, image it, take the right tissue in the right way after talking to the oncologist and the pathologist, and keep the airway safe while doing it.

AT A GLANCE

- Lymphoma is the commonest; squamous carcinoma is rare.
- Chest X-ray before anaesthesia: the mediastinal mass.
- Lymphoma node: excise it and send it fresh.
- Rhabdomyosarcoma: orbital best, parameningeal worst; the bleeding aural polyp.
- Neuroblastoma: Horner syndrome, heterochromia, raised HVA and VMA.

1 CHAPTER ONE Overview

Different tumours, different rules.

1.1 How common

Cancer is still the second commonest cause of death in childhood after trauma, and around 12% of childhood malignancies arise in the head and neck. Squamous cell carcinoma is extremely rare in children.

THE MALIGNANCIES OF THE CHILD'S HEAD AND NECK

TUMOUR	TYPICAL AGE	TYPICAL PRESENTATION
Hodgkin lymphoma	Adolescents	Firm, painless nodes, often supraclavicular; fever, night sweats, weight loss
Non-Hodgkin lymphoma	2 to 12 years	Rapidly enlarging nodes, Waldeyer's ring, often widespread
Rhabdomyosarcoma	Under 5 years in almost half	Painless mass in the orbit, nose, sinuses, nasopharynx, ear or neck

TUMOUR	TYPICAL AGE	TYPICAL PRESENTATION
Neuroblastoma	Infants, under 1 year	Firm fixed neck mass, Horner syndrome
Thyroid carcinoma	Older children and adolescents	Thyroid nodule with cervical nodes
Nasopharyngeal carcinoma	Adolescents	Neck nodes, glue ear in one ear, nasal obstruction
Salivary malignancy	Older children	Solitary parotid mass, rapid growth

WHERE THE BOOKS DIFFER

Oxford Textbook
lymphoma 27%, neural tumours 23%, thyroid tumours 21%, soft tissue sarcomas 12% of childhood head and neck malignancies.

Oxford Specialist Handbook: lymphoma about 50%, rhabdomyosarcoma 20%, neuroblastoma 6%, nasopharyngeal carcinoma 6%, thyroid carcinoma 4%; rhabdomyosarcoma and neuroblastoma are the commonest under 2. Both agree that lymphoma leads and squamous carcinoma is rare; the difference reflects the series and the age range counted.

CHAPTER TWO

2 Assessment

The lump that does not behave.

2.1 History

Duration and growth, fever, night sweats, weight loss, itching, pain, nasal obstruction or bleeding, one-sided hearing loss or ear discharge, change in the voice or breathing, swallowing, and contact with tuberculosis, travel or HIV.

2.2 Examination

Ears, nose, mouth and teeth, and the whole neck, then the axillae, groins, abdomen (liver and spleen) and skin. In the eye look for proptosis, Horner syndrome and heterochromia. Re-examine once fever, pain and dehydration have been corrected, as the signs often change.

FEATURES THAT NEED URGENT INVESTIGATION

- A node larger than about 2 cm, or one that keeps growing or has not shrunk after about 6 weeks.
- A supraclavicular node at any size.
- A hard, fixed or matted mass.
- Fever, night sweats, weight loss, pallor, bruising, or an enlarged liver or spleen.
- Stridor, dysphagia, cranial nerve palsy, Horner syndrome or proptosis.
- A one-sided nasal or nasopharyngeal mass, or an aural polyp with facial weakness.

Any large, rapidly growing or unusual lump needs urgent imaging and tissue.

2.3 Investigations

- **Ultrasound** first: it tells cystic from solid, shows the architecture of nodes and guides biopsy.
- **Chest X-ray** before any general anaesthetic, to look for a **mediastinal mass**.
- **Full blood count and film, LDH and liver function**; urine catecholamines (HVA and VMA) if neuroblastoma is possible; EBV serology in suspected nasopharyngeal carcinoma.
- **MRI** (or CT) of the primary and neck to define the extent; staging CT, bone scan, MIBG scan or bone marrow as the oncologist directs.

THE MEDIASTINAL MASS AND THE ANAESTHETIC

A child with lymphoma may have a large anterior mediastinal mass that compresses the trachea. Induction of general anaesthesia can cause complete airway or cardiovascular collapse. Get a chest X-ray first; if there is a mass, discuss with the anaesthetist and oncologist, and consider biopsy under local anaesthetic or sampling another site.

2.4 Taking tissue

Talk to the oncologist and the pathologist *before* the biopsy so that the sample answers the question.

- **Excision biopsy** of a whole node is preferred for suspected lymphoma: the architecture matters. Send it *fresh*, not in formalin, so that part can go for flow cytometry, cytogenetics and molecular tests.
- **Core biopsy** under ultrasound is an alternative if the pathologist is confident it will be diagnostic.
- **Fine-needle aspiration** is of limited use in children: it rarely gives enough for lymphoma typing.
- A deep primary (nasopharynx, sinuses, ear) is biopsied endoscopically or under anaesthetic.

OPERATIVE GUIDES IN THEATRE 3 operations

Office-Based Procedures: Fine-Needle Aspiration Biopsy of a Mass in the Head and Neck
Fine-needle aspiration

Open Thyroidectomy
Thyroidectomy

Superficial Parotidectomy
Parotidectomy

3 CHAPTER THREE The tumours

What each looks like and how it is treated.

3.1 Hodgkin lymphoma

Commonest in adolescents and young adults. Enlarged, rubbery, non-tender nodes, especially supraclavicular; up to a third have fever, night sweats, weight loss or itching. Staged by the Ann Arbor system after CT of the neck, chest, abdomen and pelvis. Treatment is chemotherapy, sometimes with radiotherapy, and the ENT role is the diagnostic biopsy. Early disease is cured in about 90%.

3.2 Non-Hodgkin lymphoma

The commonest lymphoma between 2 and 12 years. More often widespread at presentation, it may involve Waldeyer's ring (a unilateral enlarged tonsil) and grow very fast, as in Burkitt lymphoma. Immunophenotyping separates B-cell and T-cell types. Combination chemotherapy is the mainstay; prognosis is poorer than Hodgkin lymphoma but good for early disease.

THE ASYMMETRICAL TONSIL

A child with one tonsil much larger than the other, especially if it grows quickly, looks abnormal, or comes with nodes or systemic symptoms, needs the tonsil removed for histology to exclude lymphoma. Send it fresh.

3.3 Rhabdomyosarcoma

The commonest soft tissue sarcoma of childhood; about 43% occur under 5 years and the head and neck is the commonest site. Histological types are *embryonal* (about 60%, better prognosis, including the botryoid type), *alveolar* (about 20%, worse) and others. Sites are grouped as:

- **Orbital:** proptosis; the best outlook.
- **Parameningeal:** nasopharynx, sinuses, middle ear and mastoid, infratemporal fossa; spread to the skull base and meninges is common, so these are unfavourable.
- **Other head and neck sites:** face, scalp, mouth, neck, larynx.

In the ear it mimics chronic otitis media with an aural polyp, discharge, bleeding and facial palsy. Treatment is multimodal under a paediatric oncology protocol: biopsy, chemotherapy and radiotherapy, with surgery when the tumour can be removed without disabling loss of function. Five-year survival ranges from about 90% for completely resected disease to about 40% for metastatic disease.

THE AURAL POLYP IN A YOUNG CHILD

An aural polyp with bleeding, facial weakness or rapid regrowth in a young child is a rhabdomyosarcoma until proved otherwise. Send every aural polyp for histology.

3.4 Neuroblastoma

The commonest malignancy of infants under 1 year, from neural crest cells of the sympathetic chain. Only about 5% arise in the neck, where it presents as a firm, fixed, non-tender mass with ipsilateral *Horner syndrome* and *heterochromia of the iris*, and sometimes stridor, dysphagia or cranial nerve palsies. Urinary HVA and VMA are raised in about 90%. It is staged with MRI or CT, MIBG and bone scans and bone marrow biopsy. Localised disease is excised; unresectable or metastatic disease is treated with chemotherapy. Cervical neuroblastoma does better than other sites because it is noticed early, and infants under 1 often do well whatever the stage.

3.5 Thyroid carcinoma

Mostly papillary carcinoma. A thyroid nodule in a child is more likely to be malignant than one in an adult, and children more often present with cervical node and lung metastases, yet survival is excellent. Previous radiation to the neck is a risk factor. *Medullary carcinoma* occurs in multiple endocrine neoplasia type 2, where children with a RET mutation are offered prophylactic thyroidectomy. Management is by a specialist thyroid team: total thyroidectomy with the appropriate neck dissection, and radioiodine for selected patients.

3.6 Nasopharyngeal carcinoma

Rare in children, mainly in adolescents, and usually the undifferentiated type associated with Epstein-Barr virus. It presents as in adults with neck nodes, a unilateral middle ear effusion, nasal obstruction or epistaxis and cranial nerve palsies. In a child with a nasopharyngeal mass the differential is rhabdomyosarcoma, non-Hodgkin lymphoma and, in an adolescent boy, angiofibroma. Treatment is chemoradiotherapy.

3.7 Salivary gland tumours

At least two thirds of salivary swellings in children are not neoplastic, most commonly mucoceles and sialadenitis, and benign tumours are more often haemangiomas or vascular malformations than adenomas. But a *solitary solid salivary mass* in a child is malignant in more than half (against about a quarter in

adults). *Mucoepidermoid carcinoma* is the commonest malignancy (about half), followed by acinic cell carcinoma. The parotid is affected more than the submandibular gland, and rapid growth without infection suggests malignancy. Management follows adult principles: imaging, fine-needle aspiration or core biopsy, and parotidectomy with facial nerve preservation.

3.8 Other tumours

Langerhans cell histiocytosis (lytic skull and temporal bone lesions, otorrhoea, nodes), teratomas and other germ cell tumours, Ewing sarcoma, osteosarcoma, chondrosarcoma, and olfactory neuroblastoma are rare.

4 CHAPTER FOUR Treatment and after

The team and the long view.

4.1 The multidisciplinary team

Childhood cancer is treated in a paediatric oncology centre under national or international protocols, with paediatric oncologists, radiation oncologists, surgeons, pathologists, radiologists, nurses, psychologists, dietitians and speech therapists. The ENT surgeon diagnoses, secures the airway, provides surgical resection or access when needed, and manages the ear, nose and throat effects of treatment.

4.2 Late effects

Survivors live for decades with the consequences of treatment, so follow-up is long term.

- **Radiotherapy to the head and neck:** impaired growth of bone and teeth, facial asymmetry, Eustachian tube dysfunction and glue ear, sensorineural hearing loss, cataract, hypothyroidism, pituitary dysfunction, osteoradionecrosis and second cancers.

- **Cisplatin:** permanent high-frequency hearing loss, worse in young children; monitor hearing during and after treatment.
- **Surgery:** cosmetic deformity, cranial nerve deficits, swallowing and speech problems.
- **Psychological and educational** effects for the child and family.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The **commonest malignancies** in a child's neck, and that squamous carcinoma is rare.
- The **red flags** of a malignant node.
- **Chest X-ray before the anaesthetic** for a mediastinal mass.
- How to send a node for lymphoma: **excision, fresh, not in formalin**.
- **Rhabdomyosarcoma:** types, sites, parameningeal spread, the aural polyp.
- **Neuroblastoma:** Horner syndrome, heterochromia, HVA and VMA.
- Why a **solitary salivary mass** in a child is worrying.
- **Late effects** of radiotherapy and cisplatin.

5.2 Common pitfalls

COMMON PITFALLS

- Repeated courses of antibiotics for a persistent hard node.
- Putting a lymphoma node into formalin.
- Anaesthetising a child with stridor or a mediastinal mass without a chest X-ray and planning.
- Calling a bleeding aural polyp in a toddler chronic otitis media.

- Fine-needle aspiration as the only test for suspected lymphoma.

Written for this app from Oxford Textbook of Otolaryngology ch. 81, Oxford Specialist Handbook ch. 23 and K. J. Lee ch. 24, 34. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.